

GENERAL SPECIFICATION G2.2

Project Controls and Programme for the Works

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G2.2 PROJECT CONTROLS AND PROGRAMME FOR THE WORKS

G2.2.1 Overview

- (a) To ensure that information provided by the Contractor can be readily incorporated into the S.O.'s project controls systems, the S.O. requires the application of proven and consistent data structures, business processes, tools and techniques across the TWRP Project. Key to the successful delivery will be consistent and proven program controls processes, structure, and tools to disseminate and integrate timely, accurate programme and expenditure information.
- (b) This part of the specification outlines the program control discipline and management processes required for Programme for the Works, Cashflow forecast, Cost reporting, Change Management, Risk Management and Progress Reporting.

G2.2.2 Key Personnel

- (a) The Contractor's project organisation shall include the following full-time key personnel as a minimum to execute the project controls requirements. The project controls personnel shall have the experience and qualifications outlined in Volume 2.
 - (i) Project Controls Manager who shall be independent of the design and construction functions, and have an independent link to the Contractor's Project Manager. The Project Controls Manager is required to be the owner of the Contractor's Project Controls Plan and to work with the Contractor's Project Manager to implement and manage the systems, processes and people associated with the Project Controls requirements outlined in this specification and other contract documentation.
 - (ii) Senior Planner who shall maintain the Programme for the Works and related coordination and reporting requirements including Earned Value Management.
 - (iii) Project Controller(s) - Staff dedicated to Project controls shall qualified and experienced in either planning, cost control or risk management with a minimum of seven (7) years of experience in the construction industry, preferably within a site based construction management team. Project Controllers must be experienced users of Primavera P6 software and MS Excel.
 - (iv) Risk Manager who shall define the strategy and provide leadership in Risk Management for the Works and related coordination and reporting requirements including facilitating the risk review workshops.

G2.2.3 Contractor's Project Controls Plan

- (a) The Contractor shall provide a Project Controls Plan (PCP) with the initial management plan document submissions (during the contract mobilisation period); demonstrating compliance with the project controls requirements described in this document and related specifications. The PCP shall identify all systems, processes, tools and key personnel to be deployed to execute and manage the Project Controls requirements under the Contract. The PCP shall reference other project controls and management plan documents as needed.

- (b) The PCP shall describe all aspects of the Contractor's project controls functions, including resources, change management, Programme for the Works and earned value assessment with respect to project expenditure and progress, risk management, financial and progress reporting and automated systems, where applicable.
- (c) The PCP will be a live document and will be owned and produced by the Contractor's Project Controls Manager. The Contractor shall submit the PCP for the S.O.'s approval, after which the PCP shall be updated when required to reflect agreed changes as soon as practically possible.

G2.2.4 The Programme for the Works

- (a) The Programme for the Works shall be in accordance with the requirements outlined in this Specification.
- (b) The Contractor shall liaise with the S.O. within seven (7) calendar days of the Letter of Acceptance for preparation of the Programme for the Works. This will facilitate the review and acceptance plan of the Programme for the Works between the Contractor and the S.O..
- (c) Within fourteen (14) calendar days after the date of the Letter of Acceptance, the Contractor shall submit a mobilization programme indicating site arrival dates for all site personnel, temporary facilities, construction equipment, key personnel, indirect equipment and staff camps as appropriate, in support of the Programme for the Works.
- (d) The Contractor shall submit the Programme for the Works in accordance with the PSSCoC, Clause 9.1(1). The Programme for the Works shall be a development of the preliminary programme submitted during tender. The S.O. will work closely with the Contractor to agree the programme.
- (e) The Programme for the Works shall be submitted in accordance with the Information Management requirements (G1.2) and in accordance with Clause 3.2 of the PSSCoC. The Programme for the Works shall be in softcopy (.pdf and .xer) prepared using Primavera P6 software, version 15.2 or later.
- (f) The Contractor shall refer to Particular Specification P2 Preliminaries and General for the breakdown structure of the programme including milestones and hold points relevant to the Contract.
- (g) The Programme for the Works shall be submitted in accordance with the content and structure described herein.
- (h) The softcopy .pdf submission to be in bar chart / Gantt chart format and is to include minimally the following activity details:
 - (i) Activity identification number and description (in line with the breakdown structure provided for the Contract by the S.O.);
 - (ii) Original duration;
 - (iii) Early start and early finish dates;
 - (iv) Start and finish date;

- (v) Total float; and
- (vi) % complete.
- (i) The S.O. will notify the Contractor in writing of acceptance or non-acceptance of the Programme for the Works in accordance with Conditions Clauses 9.1.
- (j) Once the Programme for the Works has been accepted by the S.O., the Contractor shall not deviate from this without permission from the S.O.. For Acceptance of Programme - Refer to Conditions Clause 9.3.
- (k) The Contractor shall regularly update the Programme for the Works to reflect the actual progress of the Works. The Contractor shall submit the updated Programme for the Works with the Monthly Progress Report.
- (l) The Contractor shall provide a summary level A0 hard copy of the Programme for the Works as part of the monthly progress report for display in the project office. Detail of the summary A0 sized output to be agreed between the S.O. and the Contractor.
- (m) The Contractor shall notify the S.O. immediately of any significant impact to the Programme for the Works at any time during the execution of the Works such as but not limited to change in sequence of works proposed by the Contractor, impacts from other interfacing works or changes as a result of Variations. Any proposed revision to the Programme for the Works shall be submitted in writing with the update or as it is determined by either party. For revision of Programme - Refer to Conditions Clause 9.2.
- (n) At the end of the Contract Period, the Programme for the Works, shall reflect actual activity start and finish dates, durations and sequences of all activities.
- (o) In the event that the Programme for the Works is not acceptable to the S.O., the Contractor shall submit an acceptable Programme for the Works and the S.O. will respond in writing in accordance with Conditions Clause 9.1(3).
- (p) For failure to Submit Adequate Programme - Refer to Conditions Clause 9.4.

G2.2.5 Content and Structure of the Programme for the Works

- (a) The Programme for the Works shall be comprehensive and shall represent all activities including submittals and procurement necessary to complete the Contract. Typical minimum detail on the programme shall include the following elements:
 - (i) All major project milestones shall be identified and these shall include the interface milestones with other Contractors, Agencies, Service Providers, etc.;
 - (ii) Compilation, submission and acceptance of documentation such as management plans as part of mobilisation;
 - (iii) Site investigation works, diversion of services, temporary or permanent works;
 - (iv) Site possession requirements and Commencement of Works;
 - (v) Statutory, Licence or Permitting requirements;
 - (vi) Design activities and deliverables;
 - (vii) Submittals;
 - (viii) Key Review hold points for the S.O.;

- (ix) Periods for procuring major equipment;
 - (x) Fabrication periods at manufacturers' works;
 - (xi) Scheduled arrival of materials;
 - (xii) Soil investigation and improvement activities;
 - (xiii) Construction and installation periods;
 - (xiv) Testing and commissioning periods;
 - (xv) Interim operations and maintenance periods;
 - (xvi) Defects liability periods in Post construction condition surveys;
 - (xvii) Special operational requirements;
 - (xviii) Completion and handover of documents upon Substantial Completion including as – constructed documentation and operations / maintenance manuals;
 - (xix) Any other activities or conditions which may affect the Programme for the Work; and
 - (xx) Completion of Work.
- (b) Where required by the Contract, the Programme for the Works shall incorporate partial completion of certain sections of the Contract and shall emphasize the main activities, which are to be executed, ensuring the completion of the entire works within the stipulated contract period.
- (c) The Contractor shall indicate clearly in the Programme for the Works the activities and periods where night work and 24 hours working are required to achieve the key milestones. Night work and 24 hours working will require prior approval from the S.O..
- (d) The Programme for the Works shall include resource loading / as required to establish histograms and S-curves (i.e. manpower, construction equipment and material resources available).
- (e) The Programme for the Works shall include a cashflow S-curve based on projected monthly payments and milestone payments as specified in the Contract. The programme to be expenditure loaded to match requirement described in G2.2.10.
- (f) The activities shall not contain any start-finish relationship, and should not contain start-start relationship unless written agreement with the S.O. is obtained.
- (g) A breakdown structure shall be developed by the Contractor in line with the high-level structure and activity codes referenced in the Particular Specification P2 Preliminaries and General. The Contractor to develop further levels of breakdown in consultation with and subject to acceptance by the S.O., to ensure compatibility with the overall project-wide breakdown structure and BIM reporting requirements. The Programme to be expenditure loaded to this level as a minimum. The Contractor shall structure and code the Programme for the Works to support the lowest level b breakdown elements of the works. These same activities in the Programme for the Works can then be summarised using the breakdown hierarchy.
- (h) A design deliverables register that reflects the programmed design activities in more detail shall be established by the Contractor for accurate progress monitoring purposes and reporting.

- (i) Procurement of materials and equipment shall be reported in terms of percent (%) complete. The Contractor shall provide a procurement register showing each intended procurement item, vendor/manufacturer, approval dates by S.O., actual purchase order date and float to criticality. The procurement register for equipment shall indicate at minimum, Tender Calling; Tender Close; Tender Evaluation; Contract Award; Vital Vendor Information for engineering design; Vendor Work commencement; Vendor Factory Acceptance Test (FAT); Vendor delivery; Vendor final documentation. Planned, Actual and Forecast dates for each item to be clearly shown. This register shall be used as the basis to derive the % complete. The procurement register for bulk material shall at minimum be broken down by individual batch delivery items including the information on quantities expected to be delivered.
- (j) Construction progress is expressed in terms of Percent (%) complete and is quantitatively derived from the actual work completed compared to baseline planned work required for the completion of an activity.
- (k) The Contractor shall ensure that all activities in the Programme for the Works, which are associated with milestone lump sum payments, have performance measurement techniques identified which will show how progress will be measured. The Contractor shall submit a guideline document outlining these techniques for S.O.'s approval. The document shall explain clearly how the weightage of different scope items is defined as it may vary for different activities e.g. piling may be measured by numbers of piles, tunnelling may be measured by metres tunnelled. This also applies to detailed design work carried out by the Contractor which shall be expressed in terms of Percent (%) Complete.
- (l) The Programme for the Works shall be in the form of a Critical Path Network. The Critical Path will be identified as the longest path.
- (m) Negative lags on activity relationships shall not be accepted.
- (n) All key milestones, constraints, activity relationships, dependencies and float shall be shown.
- (o) The duration of any activity shall not exceed 30 working days, except where otherwise agreed by the S.O..
- (p) The calendar used shall be based on a 7 day work week with no holidays. Any non-work restriction shall be deemed to have been considered in the calendar to plan the Programme the Works. The time unit shall be in days. Date format shall be DD/MM/YY. Furthermore, the calendars within the Programme for the Works are to be set on "8 work hours/ day". These requirements are only so that the Programme for the Works is consistent with all other DTSS2 contracts and is purely a technical P6 requirement. This is not a restriction placed on the planned working hours for the actual works on Site, nor does it require the Contractor to work 7 days/week unless proposed by the Contractor and approved by the S.O..
- (q) If the actual work activity on Site, represented in the Programme for the Works, is based on hours greater than or less than 8 hours per day, then the programme calendar description shall provide a comment to explain why that is the case.

- (r) As far as the Programme for the Works is concerned, a week shall be assumed to begin at 0000 hours on a Monday and ends at 2359 hours on a Sunday. The completion of an activity or achievement of any event when given a week number shall be taken to mean midnight on the Sunday at the end of the numbered week. An access date or activity start date when given as week number shall be taken to mean 0000 hours on a Monday of the numbered week.
- (s) The Contractor shall allow ample float time for consultation with and obtaining clearance from, property owners and other any authorities departments/ agencies.
- (t) The Contractor shall take in to consideration any limitation in working space and existence of other installations and services within and around the site in preparation of Programme for the Works.
- (u) The Contractor shall incorporate activities and the time frame for carrying out services detection for the Site.
- (v) When the S.O. considers it necessary for the Contractor to divert existing services, the Contractor shall add this detail to the Programme for the Works. The Contractor shall note that not all services encountered will need to be diverted. Where services are to be diverted, the Contractor shall allow sufficient float for any approvals by authorities to carry out the diversion works.
- (w) Contractor shall provide a slippage report to the S.O. with each submission of the monthly Programme for the Works update. Planned progress shall be generated from Primavera P6, and earned progress shall be generated by measuring physical work completed. Activities behind planned progress shall be highlighted in this report.
- (x) The complete scope of work shall be covered in the Programme for the Works. Activities shall be logically connected in a manner which reflects the interdependencies at the detail level, as far as is practical.
- (y) Notwithstanding the above, each activity in the Programme for the Works shall map to one item in the Bill of Quantities to enable the Contractor to summarise percent complete against the bill items for monthly progress claims.

G2.2.6 Linking the Programme for the Works with the BIM Model (4D)

- (a) The Contractor shall ensure the activities are sufficiently detailed in the Programme for the Works (Primavera P6) to allow mapping to activity codes in the BIM model (4D) to show planned and actual construction sequences. This shall be completed as a six-monthly forecast and updated monthly.
- (b) This shall be done by assigning a set of codes (4D attributes) to the activities in the Programme for the Works and exporting these codes from the P6 file (as an Excel output or similar) to be embedded into the BIM model. The same codes shall be embedded within the “4D Sequence Code” Attribute within the BIM model such that a match between a single Activity in the programme can reference multiple elements in the BIM model. Refer to General Specifications G1.2 - Information Management for further information on BIM.
- (c) Guidance on the TWRP 4D Sequence (codes) will be provided to the Contractor by the S.O. upon Contract award.

G2.2.7 Narrative of the Basis of the Programme for the Works

- (a) The Contractor shall submit a written narrative basis of programme containing information of calendar(s) used for activities, assumptions, build methodology, norms and productivity, duration computation, critical path analysis, list of constraints used with the explanation why it is used, manpower histograms, resources planned, progress S-curves.

G2.2.8 Programme Coordination Requirements

- (a) The Contractor shall meet with the S.O. twice every month to review and update the detail of the Programme for the Works. The first meeting will be mid-month and will focus on the forecast to the end of the month; whereas the month-end meeting will compare monthly actual and planned progress. The meetings will review interface milestone dates, progress status, remaining duration for ongoing activities, 3 months lookahead at the minimum. The Contractor is expected to submit a softcopy (.pdf and .xer) of the Programme for the Works to the S.O. minimum two days prior to these coordination meetings.
- (b) The Contractor shall also attend a monthly interface meeting, convened by the S.O. with all other TWRP contractors and any other stakeholders deemed necessary by the S.O.. The meeting will be chaired by the S.O. and will focus on interface coordination milestones between contractors.
- (c) From time to time the S.O. may require the Contractor's attendance at further coordination meetings to review and update the Programme for the Works.
- (d) A full calendar of meetings and reporting requirements will be made available to the Contractor after award.

G2.2.9 Programme for the Works - Updates and Performance Measurement

- (a) The monthly updating of the Programme for the Works is necessary to verify the Contractor's progress claims. If the Contractor fails to provide sufficient information for the S.O. to approve a progress claim, the Contractor shall not be entitled to progress payment until the necessary information is furnished and the S.O.'s approval is obtained.
- (b) The Contractor's monthly update of the Programme for the Works and outputs described herein are to be submitted as part of the suite of documents accompanying the Monthly Progress Report.
- (c) The updates to the Programme for the Works shall contain as a minimum complete information on the following items:
 - (i) Critical path;
 - (ii) Actual completion dates for completed activities;
 - (iii) Remaining duration required to complete each activity started, or scheduled to start, but not completed;
 - (iv) Previous month percent complete and current percent complete;
 - (v) Baseline start and finish dates;
 - (vi) Gantt charts that reflects baseline, actual and remaining;

- (vii) Individual and total float; and
- (viii) Details of any changes in programme logic and timing required due to factors either within or outside the control of the Contractor.
- (d) The actual % complete against baseline planned % shall be presented in table and S-curve format.
- (e) Key Performance Indicators (KPI) shall be utilised in addition to the above progress measurements; such as Schedule Performance Index (SPI).
- (f) Further KPI based on major resources and/or commodities shall be developed by the Contractor for review and acceptance by the S.O.. The KPI shall be presented in table, histogram and S-curve format.
- (g) The Contractor shall submit a narrative report following the month end programme coordination meeting, in a form agreed between by the Contractor and the S.O.. The narrative (a short report) shall include a description of problem areas; changes instructed, current and anticipated delaying factors and their estimated impact on performance of other activities and completion date; and an explanation of corrective action taken or proposed. This will form part of the monthly progress report suite of submissions.
- (h) Reporting format and content for Programme for the Works shall be developed by the Contractor in consultation with and subject to acceptance by the S.O. to ensure compatibility with the overall project reporting requirements (refer to Monthly Progress Reporting requirements).
- (i) The month end date shall be determined as the last day of each calendar month.

G2.2.10 Project Expenditure Management, BIM (5D) and Cashflow Forecasts

- (a) The Contractor shall utilize the Model for Cost estimating, cash flow forecasting and reporting on actual expenditure. The design intent model has sufficient classification to enable exporting of the underlying data for Cost estimation purposes.
- (b) A breakdown structure shall be developed by the Contractor in line with the Bill (BoQ) to report project expenditure. The Contractor shall develop a structure in consultation with and subject to acceptance by the S.O., to ensure compatibility with the overall project-wide breakdown structure and BIM reporting requirements. The Contractor shall structure and code the cost to the lowest level breakdown elements of the works. These lowest levels can then be summarised using the breakdown hierarchy to the BoQ for progress claims and to the BIM model.
- (c) The Contractor shall ensure project expenditure elements are sufficiently detailed so that it is suitable to link line items to element(s) in the BIM model (5D). Each model element shall be mapped to the Bill of Quantities. Any lump sum item in the BoQ may need to break down to further elements to map to an element in the BIM model.
- (d) The Programme for the Works will be the basis of the Board's planned project expenditure. The Contractor shall apportion the cost to each activity in the programme submitted for acceptance to enable measurement of monthly progress and payment.
- (e) The cashflow forecast shall be submitted in accordance with the document transmittal requirements specified in the document G2.2.

- (f) The Contractor shall ensure project expenditure elements are sufficiently detailed so that it is suitable to link items to the BIM model (5D) to show cost attributes required for the BIM model. Each BIM item needs to be mapped to the Bill of Quantities. Any lump sum item in the BoQ may need to break down to further elements to map to an item in the BIM model.
- (g) A set of codes (5D attributes) assigned to the cost elements in the expenditure file (as an Excel output or similar) need to be embedded into the “5D Code” attribute in the BIM model by the Contractor’s BIM Manager.
- (h) Guidance on the TWRP 5D attributes (codes) will be provided to the Contractor by the S.O. upon contract award.

G2.2.11 Change Management

- (a) As part of a robust project controls and management system the Contractor shall maintain a Change control process to manage the scope of works by implementing the following:
 - (i) Adhere to Clause 19 of Conditions in managing and reporting variations;
 - (ii) Programme for the Works changes in P6 to be included in the Contractor's P6 Schedule Change Log;
 - (iii) Develop and maintain a change variation log for review during monthly progress meetings; and
 - (iv) Develop a process of how the Contractor will liaise with the S.O. in order to incorporate ‘change’ into contract sum, scope, model and programme, in a timely manner.
- (b) The Contractor shall define a change control process in coordination with the S.O. which incorporates meetings, governance process and documentation in the Project Controls Plan.

G2.2.12 Risk Management

- (a) The Board and S.O. recognises that managing risk is critical to the successful delivery of the Project. The S.O. defines risk as a potential threat or opportunity of any type (e.g. programme, safety, environmental, capital expenditure or quality) which could adversely or positively affect the Contract and project milestones.
- (b) To meet the project risk management requirements, the Contractor shall:
 - (i) Ensure risk management is an integral part of the day-to-day management of the Project;
 - (ii) Improve awareness and understanding of risk and the need for effective risk management;
 - (iii) Integrate and embed risk management into the culture of the Project and provide a resource to manage risk;
 - (iv) Ensure the involvement and participation of all by providing risk management training; and
 - (v) Ensure continual improvement.

- (c) The Contractor shall develop, employ and submit to the S.O. for acceptance within thirty (30) days of the letter of acceptance, a risk management plan outlining:
 - (i) Developing, implementing and adhering to accepted risk management procedures;
 - (ii) Incorporating risk management into all areas of the Project;
 - (iii) Risk identification, reviewing and reporting;
 - (iv) Describe how he will manage contract risk with his subcontractors and supply chain; and
 - (v) Maintain a risk register with impact and probability together with mitigations.
- (d) The Contractor shall refer to the Risk Register developed by the S.O. which is referenced in Particular Specification P2 Preliminaries and General and develop a comprehensive Risk Register with proposed mitigation measures to eliminate or reduce the risks identified. A Risk Management facilitator appointed by the Contractor will be required to take a lead role in the risk management process.
- (e) The Contractor will be required to maintain and update the risk register for the duration of the Contract. The Contractor will be required to issue an updated Risk Register to the S.O. every month as part of the monthly progress report. The monthly report to summarise top ten risks as a multiple of value and probability together with mitigation plans and strategies.
- (f) The Contractor shall make his risk registers available to the Board's Insurance Broker when required by the S.O.. The Contractor, the Broker, the Board and the S.O. will jointly review the Contractor's Risk Management processes.
- (g) The Contractor shall meet at least once every month or more often if decided by the S.O. with all project stakeholders, including the first level sub-contractors, to identify risks, assign probabilities and impacts, identify mitigation measures and assign ownership of each risk. The Contractor shall provide a report to the S.O. on the outcome of each meeting.
- (h) The S.O. will set up a series of review meetings, expected to be once every three months, to monitor at a senior management level the risks for the TWRP project. The Contractor will send his Project Director, if required, to this meeting to report on the current key risks related to the works under the contract.
- (i) The Contractor shall carry out quantitative schedule risk analysis (QSRA) on the Programme for the Works to assess the confidence levels of achieving key milestones on a quarterly basis or when there has been significant change to the works.
- (j) The Contractor shall also provide a cover note that outlines the assumptions made in carrying out the QSRA, the modelling parameters, along with details of duration uncertainties for each activity that have been applied in the risk analysis.
- (k) The assumptions made in carrying out the QSRA, the modelling parameters, along with details of any duration uncertainties shall not be changed against any activity in subsequent revised programmes unless otherwise agreed with the S.O..
- (l) The Contractor to give an early warning by notifying the S.O. using a formal contract communication as soon as they become aware of any matter which could, but not limited to:

- (i) Increase the total contract value;
 - (ii) Delay Completion;
 - (iii) Delay meeting a Key Date;
 - (iv) Impair the performance of the works in use; or
 - (v) Any other matter that could impact the delivery of the TWRP project.
- (m) Once an early warning notification has been issued by the Contractor, the S.O. will instruct the Contractor to attend a risk reduction meeting. The risk reduction meeting will consider proposals on how to avoid or reduce the effect of the registered risks. The meeting is to seek solutions that will bring advantage to all those who will be affected and decide upon actions and who, in accordance with the terms of the contract, will action them. This meeting may also review which risks have now been avoided or have passed and can be removed from the Early Warning Register.

G2.2.13 Reporting Earned Value

- (a) The Contractor shall measure project performance internally using earned value analysis (EVA) techniques and shall reflect this in the monthly report submitted to the S.O.. This is to provide confidence to the S.O. that the Contractor performance is on track to meet the overall project milestones. The performance measurement may include the integrated reporting of:
- (i) Programme;
 - (ii) Project expenditure;
 - (iii) Construction quantities; and
 - (iv) Expended man hours.
- (b) Templates for earned value reporting will be provided to the Contractor by the S.O. after award. The document will include definition of terms, performance measurement and analysis reports.

END OF SECTION